

Week 7 Activity Log

Task 3 - React Countries App & Final Project Build (Phase 1)

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Internship Program: CoderAxo Web Development Internship

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Week Duration: July 27 - August 2, 2026

Week Overview

This week ran two tracks at once: finishing Task 3, a complete deployed React application, and starting real development on the Final Project. Balancing both meant longer days than usual, but finishing Task 3 early in the week freed up the rest of it for the Final Project's first and most critical module — Authentication.

Objectives for the Week

- Complete and deploy Task 3 with real API data and full client-side routing
- Get the Final Project's Authentication module fully working end to end before moving to any other module

Daily Activity Breakdown

Mon, Jul 27

Task 3 assigned: build and deploy a full React app with functional routing, real API data, and Tailwind styling. Planned the Countries Explorer app's pages.

Tue, Jul 28

Built the Home page with a country grid, fetching live country data and rendering cards with search/filter.

Wed, Jul 29

Built the Country detail page (dynamic route), the About page, the Contact page, and a custom 404/Error page, wired together with React Router.

Thu, Jul 30

Styled the entire app with Tailwind CSS, made it responsive, deployed it, and submitted Task 3 with the live link and source code.

Fri, Jul 31

Started Final Project development. Set up the MERN project structure — Express + MongoDB Atlas backend, React + Vite frontend — using the vertical-slice plan from last week.

Sat, Aug 1

Built the Authentication backend: user model, register/login routes, JWT token generation, and bcrypt password hashing.

Sun, Aug 2

Built the Authentication frontend: Register and Login pages, an AuthContext for global auth state, and a protected Dashboard route that redirects unauthenticated users.

Detailed Summary

Task 3 was the first time I fetched and rendered real, live API data in React rather than static/mock data, which meant properly handling loading and error states instead of assuming the data would just be there. That experience carried directly into the Final Project's Auth module — fetching, loading states, and error handling all showed up again immediately when wiring the frontend AuthContext to the backend's login/register endpoints. Getting Authentication fully working (hashing, token issuing, protected routes, and a working AuthContext) before touching any other module meant the rest of the Final Project had a stable, tested foundation to build on rather than starting from an untested guess.

Tools & Technologies Used

- React 18, React Router, Tailwind CSS (Task 3)
- Express.js, MongoDB Atlas, Mongoose (Final Project backend)
- JWT, bcrypt.js (authentication)
- Vercel/Netlify (Task 3 deployment)

Skills & Concepts Learned

- Fetching and rendering real API data in React with proper loading/error handling
- Structuring a multi-page React app cleanly with nested routing
- Implementing JWT-based authentication end to end (hashing, token signing, verifying, protected routes)
- Sharing auth state across a React app using Context instead of prop drilling

Challenges Faced & How I Resolved Them

Deploying Task 3 initially failed because of an environment/build configuration issue specific to the hosting platform — routes worked locally but returned 404s on refresh once deployed, which turned out to need a redirect rule for client-side routing. On the Final Project side, the first version of the protected route redirected users even while the auth check was still loading, briefly flashing the login page for already-logged-in users; adding a loading state to the AuthContext before rendering any route fixed it.

Reflection

Running Task 3 and the start of the Final Project in the same week was the busiest stretch of the internship so far, but finishing Authentication cleanly by Sunday night meant heading into Week 8 with real momentum instead of a half-built foundation.

Goals for Next Week

- Build out the Dashboard, Team Management, Project, and Task Board modules on top of the completed Auth system

- Keep following the vertical-slice approach — one complete module before starting the next
- Leave enough time before August 5 for testing, documentation, and the demo video